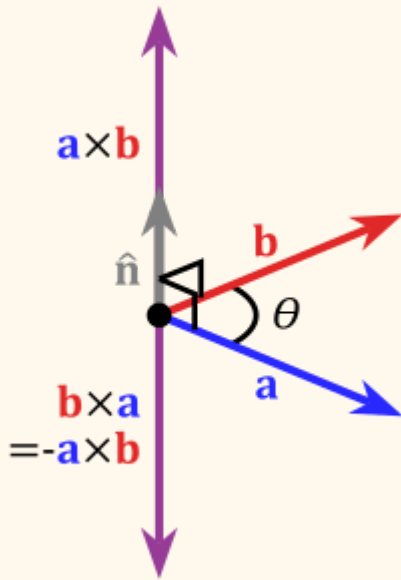


Vector Stuff

Dot and Cross Products



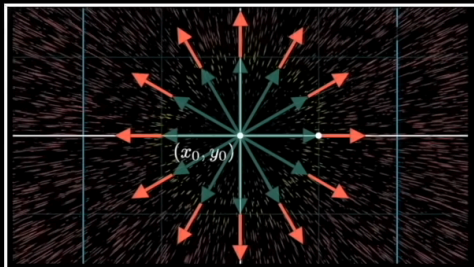
$$a \cdot b = |a||b| \cos\theta$$

$$|a \times b| = |a||b| \sin\theta$$

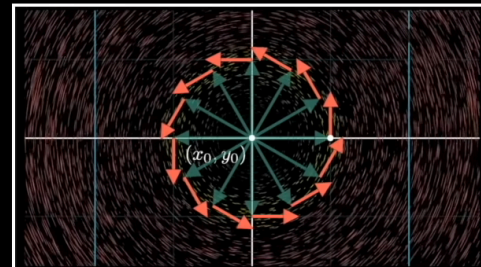
$$a \times b = |a||b| \sin\theta \hat{n}$$

Vector Operations

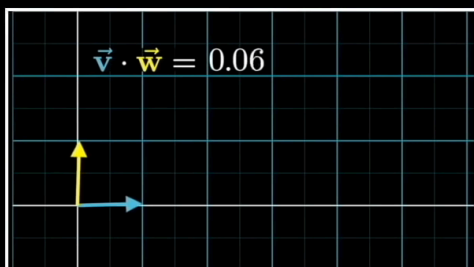
Divergence



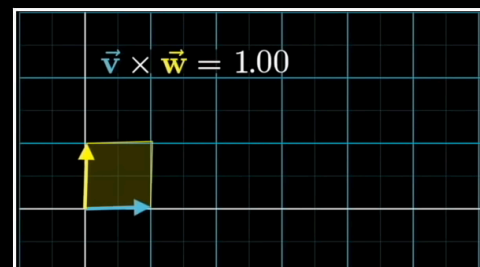
Curl



Dot product



Cross product

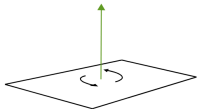


📺 Divergence and curl: The language of Maxwell's equations, fluid flow, and more

- for small step away from a point, if the change in direction is align with the step, then dot product is high, divergence/convergence
- for small step away from a point, if the change in direction is orthogonal with the step, then cross product is high, curl

Operation

Stuff

Del	$\nabla = \left(\frac{\partial}{\partial x}, \frac{\partial}{\partial y}, \frac{\partial}{\partial z} \right) = \frac{\partial}{\partial x} \mathbf{a}_x + \frac{\partial}{\partial y} \mathbf{a}_y + \frac{\partial}{\partial z} \mathbf{a}_z$
Divergence	Measure of how the field is spreading out at a point For scalar function f, the resultant is a vector field called gradient field (if the divergence is taken again, it will be the Laplacian) $\nabla f = \left(\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}, \frac{\partial f}{\partial z} \right) = \frac{\partial f}{\partial x} \mathbf{a}_x + \frac{\partial f}{\partial y} \mathbf{a}_y + \frac{\partial f}{\partial z} \mathbf{a}_z$ For vector valued function F, result of operation is a scalar function $\nabla \cdot F = \left(\frac{\partial}{\partial x}, \frac{\partial}{\partial y}, \frac{\partial}{\partial z} \right) \cdot (F_x, F_y, F_z) = \frac{\partial F_x}{\partial x} + \frac{\partial F_y}{\partial y} + \frac{\partial F_z}{\partial z}$ • $\text{div } F > 0$: sourcing • $\text{div } F = 0$: none • $\text{div } F < 0$: sinking Divergence of curl is zero [edit] The divergence of the curl of any continuously twice-differentiable vector field $\mathbf{A}$ is always zero: $\nabla \cdot (\nabla \times \mathbf{A}) = 0$ This is a special case of the vanishing of the square of the exterior derivative in the De Rham chain complex. Divergence of gradient is Laplacian [edit] The Laplacian of a scalar field is the divergence of its gradient: $\Delta \psi = \nabla^2 \psi = \nabla \cdot (\nabla \psi)$ The result is a scalar quantity. Divergence of divergence is not defined [edit] The divergence of a vector field $\mathbf{A}$ is a scalar, and the divergence of a scalar quantity is undefined. Therefore, $\nabla \cdot (\nabla \cdot \mathbf{A}) \text{ is undefined.}$
Curl	Measure of how the field is rotating about a point, result is vector valued function with direction = axis of rotation and magnitude proportional to rotational speed $\text{curl}(F) = \nabla \times F$ Rotation in three dimensions is described with a single vector.   Right hand rule to determine the direction of a rotation vector. Or just a thumb up. • $\text{curl } F > 0$: Anti Clockwise around the vector direction • $\text{curl } F = 0$: none • $\text{curl } F < 0$: Clockwise around the vector direction • when curl of a vector field = some other vector, means the vector field is circulating around the other vector

Curl of gradient is zero [edit]

The curl of the gradient of any continuously twice-differentiable scalar field φ (i.e., differentiability class C^2) is always the zero vector:

$$\nabla \times (\nabla \varphi) = \mathbf{0}.$$

It can be easily proved by expressing $\nabla \times (\nabla \varphi)$ in a Cartesian coordinate system with Schwarz's theorem (also called Clairaut's theorem on equality of mixed partials). This result is a special case of the vanishing of the square of the exterior derivative in the De Rham chain complex.

Curl of curl [edit]

$$\nabla \times (\nabla \times \mathbf{A}) = \nabla(\nabla \cdot \mathbf{A}) - \nabla^2 \mathbf{A}$$

Here ∇^2 is the vector Laplacian operating on the vector field $\mathbf{A}$.

Curl of divergence is not defined [edit]

The divergence of a vector field $\mathbf{A}$ is a scalar, and the curl of a scalar quantity is undefined. Therefore,

$$\nabla \times (\nabla \cdot \mathbf{A}) \text{ is undefined.}$$

- Curl of curl of $\mathbf{A}$ = gradient of divergence of $\mathbf{A}$ - Laplacian of $\mathbf{A}$
- not proof, but some intuition on why curl of gradient = 0
 - given a 3D surface, a scalar function V , the ∇V will be the gradient field of V
 - if ∇V is not curl free, means there is a point on the surface where by a ball can fall in circles due to the surface it self
 - this is not possible as a surface cannot exert any force on the ball as its a scalar function not vector

Laplacian

Laplacian of a scalar is the divergence of the gradient of the scalar = divergence of a gradient field of the scalar, result is a scalar function:

$$\nabla^2 f = \frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2} + \frac{\partial^2 f}{\partial z^2}$$

Laplacian of a vector when computed in orthonormal Cartesian coordinates, is equal to the vector field of the scalar Laplacian applied to each vector component, result is a vector function:

$$\nabla^2 \mathbf{F} = (\nabla^2 F_x, \nabla^2 F_y, \nabla^2 F_z)$$

$$\nabla^2 \mathbf{F} = \left(\frac{\partial^2 F_x}{\partial x^2} + \frac{\partial^2 F_x}{\partial y^2} + \frac{\partial^2 F_x}{\partial z^2}, \frac{\partial^2 F_y}{\partial x^2} + \frac{\partial^2 F_y}{\partial y^2} + \frac{\partial^2 F_y}{\partial z^2}, \frac{\partial^2 F_z}{\partial x^2} + \frac{\partial^2 F_z}{\partial y^2} + \frac{\partial^2 F_z}{\partial z^2} \right)$$

<https://physics.stackexchange.com/questions/220400/why-can-the-divergence-of-vector-potential-be-anything>

https://ccom.ucsd.edu/~ctee/notes/grad_n_curl.pdf

https://en.wikipedia.org/wiki/Vector_calculus_identities

$$\nabla(f + g) = \nabla f + \nabla g, \quad \nabla \cdot (\mathbf{A} + \mathbf{B}) = (\nabla \cdot \mathbf{A}) + (\nabla \cdot \mathbf{B}),$$

$$\nabla \times (\mathbf{A} + \mathbf{B}) = (\nabla \times \mathbf{A}) + (\nabla \times \mathbf{B}),$$

and

$$\nabla(kf) = k\nabla f, \quad \nabla \cdot (k\mathbf{A}) = k(\nabla \cdot \mathbf{A}), \quad \nabla \times (k\mathbf{A}) = k(\nabla \times \mathbf{A}),$$

- f, g are scalar functions, $\mathbf{A}, \mathbf{B}$ are vector functions, k is constants

Accordingly, there are *six* product rules, two for gradients:

$$(i) \quad \nabla(fg) = f\nabla g + g\nabla f,$$

$$(ii) \quad \nabla(\mathbf{A} \cdot \mathbf{B}) = \mathbf{A} \times (\nabla \times \mathbf{B}) + \mathbf{B} \times (\nabla \times \mathbf{A}) + (\mathbf{A} \cdot \nabla)\mathbf{B} + (\mathbf{B} \cdot \nabla)\mathbf{A},$$

two for divergences:

$$(iii) \quad \nabla \cdot (f\mathbf{A}) = f(\nabla \cdot \mathbf{A}) + \mathbf{A} \cdot (\nabla f),$$

$$(iv) \quad \nabla \cdot (\mathbf{A} \times \mathbf{B}) = \mathbf{B} \cdot (\nabla \times \mathbf{A}) - \mathbf{A} \cdot (\nabla \times \mathbf{B}),$$

and two for curls:

$$(v) \quad \nabla \times (f\mathbf{A}) = f(\nabla \times \mathbf{A}) - \mathbf{A} \times (\nabla f),$$

$$(vi) \quad \nabla \times (\mathbf{A} \times \mathbf{B}) = (\mathbf{B} \cdot \nabla)\mathbf{A} - (\mathbf{A} \cdot \nabla)\mathbf{B} + \mathbf{A}(\nabla \cdot \mathbf{B}) - \mathbf{B}(\nabla \cdot \mathbf{A}).$$

- f, g are scalar functions, $\mathbf{A}, \mathbf{B}$ are vector functions

Formula

Triple Products

$$(1) \quad \mathbf{A} \cdot (\mathbf{B} \times \mathbf{C}) = \mathbf{B} \cdot (\mathbf{C} \times \mathbf{A}) = \mathbf{C} \cdot (\mathbf{A} \times \mathbf{B})$$

$$(2) \quad \mathbf{A} \times (\mathbf{B} \times \mathbf{C}) = \mathbf{B}(\mathbf{A} \cdot \mathbf{C}) - \mathbf{C}(\mathbf{A} \cdot \mathbf{B})$$

Product Rules

$$(3) \quad \nabla(fg) = f(\nabla g) + g(\nabla f)$$

$$(4) \quad \nabla(\mathbf{A} \cdot \mathbf{B}) = \mathbf{A} \times (\nabla \times \mathbf{B}) + \mathbf{B} \times (\nabla \times \mathbf{A}) + (\mathbf{A} \cdot \nabla)\mathbf{B} + (\mathbf{B} \cdot \nabla)\mathbf{A}$$

$$(5) \quad \nabla \cdot (f\mathbf{A}) = f(\nabla \cdot \mathbf{A}) + \mathbf{A} \cdot (\nabla f)$$

$$(6) \quad \nabla \cdot (\mathbf{A} \times \mathbf{B}) = \mathbf{B} \cdot (\nabla \times \mathbf{A}) - \mathbf{A} \cdot (\nabla \times \mathbf{B})$$

$$(7) \quad \nabla \times (f\mathbf{A}) = f(\nabla \times \mathbf{A}) - \mathbf{A} \times (\nabla f)$$

$$(8) \quad \nabla \times (\mathbf{A} \times \mathbf{B}) = (\mathbf{B} \cdot \nabla)\mathbf{A} - (\mathbf{A} \cdot \nabla)\mathbf{B} + \mathbf{A}(\nabla \cdot \mathbf{B}) - \mathbf{B}(\nabla \cdot \mathbf{A})$$

Second Derivatives

$$(9) \quad \nabla \cdot (\nabla \times \mathbf{A}) = 0$$

$$(10) \quad \nabla \times (\nabla f) = 0$$

$$(11) \quad \nabla \times (\nabla \times \mathbf{A}) = \nabla(\nabla \cdot \mathbf{A}) - \nabla^2 \mathbf{A}$$